Day 2 Hands-on:
Forces and phonons

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Today's Themes:

Phonons: compute phonons in Γ and along the BZ, for Si and AlAs

Exercise 1.a:

Phonons at Γ in non-polar materials

Phonons at Γ in non-polar materials

Go to the directory with the input files:

cd into 01_Si_QE_a

In this directory you will find:

- *README.md* – File describing how to do the exercise
- *Si.scf.in* – Input file for the SCF ground-state calculation
- *Si.ph.in* – Input file for the phonon calculation at Γ
- *Si.dynmat.in* – Input file to impose the acoustic sum rule
- **reference** – Directory with the reference results

Phonons at Γ in non-polar materials

```
&control
   calculation='scf'
   restart_mode='from_scratch',
   pseudo_dir='../../files/pseudo'
   outdir='./out'
   prefix='Si',
 /
 &system
   ibrav = 2,
   celldm(1) = 10.20,
   nat = 2,
   ntyp = 1,
   ecutwfc = 16
 /
 &electrons
   diagonalization = 'davidson'
   mixing_mode = 'plain',
   conv_thr =  1.0d-8
   mixing_beta = 0.7
 /
ATOMIC_SPECIES
 Si  28.0855 Si.vbc.UPF
ATOMIC_POSITIONS
 Si 0.00 0.00 0.00
 Si 0.25 0.25 0.25
```

Step 1: Perform a Self-Consistent Field ground-state calculation for silicon at the equilibrium structure using the pw.x program.

```
mpirun –np 2 pw.x –in Si.scf.in > Si.scf.out
```

Input Output

Should generally be smaller than for simple total energy!

Phonons at Γ in non-polar materials

Input file for phonon at Gamma

```
Phonons at Gamma
 &inputph
  prefix = 'Si',
  tr2_ph = 1.0d-14,
  amass(1) = 28.0855,

  outdir = './out'
  fildyn = 'Si.dyn',
 /
 0.0  0.0  0.0
```

Step 2: Perform a phonon calculation at Γ using the ph.x program.

```
mpirun –np 2 ph.x –in Si.ph.in > Si.ph.out
```

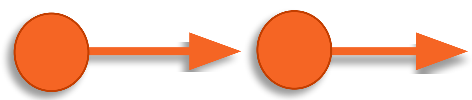
- prefix: The same prefix as in the SCF calculation
- tr2_ph: Threshold for self-consistency
- amass(1): Atomic mass
- outdir: Directory for temporary files
- fildyn: File containing the dynamical matrix
- 0.0 0.0 0.0: Coordinates of the q point (Γ) in units of 2*pi/a in the Cartesian reference system

Phonons at Γ in non-polar materials

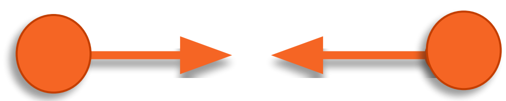
Dynamical matrix file: **Si.dyn**

```
     Diagonalizing the dynamical matrix

     q = (    0.000000000   0.000000000   0.000000000 )

 **************************************************************************
     freq (    1) =       0.369224 [THz] =      12.315990 [cm-1]
 ( -0.502437  0.000000  0.469222  0.000000  0.165492  0.000000 )
 ( -0.502437  0.000000  0.469222  0.000000  0.165492  0.000000 )
     freq (    2) =       0.369224 [THz] =      12.315990 [cm-1]
 ( -0.186720  0.000000  0.040185  0.000000 -0.680824  0.000000 )
 ( -0.186720  0.000000  0.040185  0.000000 -0.680824  0.000000 )
     freq (    3) =       0.369224 [THz] =      12.315990 [cm-1]
 (  0.461186  0.000000  0.527462  0.000000 -0.095350  0.000000 )
 (  0.461186  0.000000  0.527462  0.000000 -0.095350  0.000000 )
     freq (    4) =      15.306901 [THz] =     510.583264 [cm-1]
 (  0.700260 -0.000000 -0.059885  0.000000 -0.077779  0.000000 )
 ( -0.700260  0.000000  0.059885 -0.000000  0.077779  0.000000 )
     freq (    5) =      15.306901 [THz] =     510.583264 [cm-1]
 (  0.067676  0.000000  0.700365  0.000000  0.070066  0.000000 )
 ( -0.067676  0.000000 -0.700365  0.000000 -0.070066  0.000000 )
     freq (    6) =      15.306901 [THz] =     510.583264 [cm-1]
 ( -0.071103  0.000000  0.076832  0.000000 -0.699315  0.000000 )
 (  0.071103  0.000000 -0.076832  0.000000  0.699315  0.000000 )
 **************************************************************************
```

Acoustic modes (in-phase)



Optical modes (out-of-phase)



Phonons at Γ in non-polar materials

Acoustic sum rule at Γ

Problems with the frequency of the acoustic phonon mode at Γ and with effective charges.
Because of the numerical inaccuracies the interatomic force constants (IFC) and effective charges (Z*) do not strictly satisfy the acoustic sum rule (ASR).

ASR comes directly from the continuous translational invariance of the crystal:
if we translate the whole solid by a uniform displacement, the forces acting on the atoms must be zero

$\Longrightarrow$ for each cartesian direction α, β and *s*,*s'*-th atom:

$$\sum_{\boldsymbol{R}j} C_{s\alpha,s'\beta}(\boldsymbol{R}) = 0, \qquad \sum_{j} Z^{*}_{s\alpha\beta} = 0$$

As a consequence, the frequencies of the acoustic modes at Gamma must be zero.

Phonons at Γ in non-polar materials

Acoustic sum rule at Γ

Reason for numerical inaccuracy:

- Insufficiently accurate SCF thresholds (in pw.x and/or ph.x)
- XC energy is computed in real space. More problematic for GGA than in LDA, for US pseudopotentials it could require large *ecutrho*.
- K-points sampling not accurate enough (especially, BECs and dielectric constant required denser k-point sampling).

Phonons at Γ in non-polar materials

Acoustic sum rule at Γ

The acoustic sum rule (ASR) can however be imposed after the phonon calculations. To do this we use the dynmat.x program that imposes the ASR on the elements of the dynamical matrix and diagonalize it.

Step 3: Run a dynmat.x calculation to impose the ASR

```
mpirun –np 2 dynmat.x –in Si.dynmat.in > Si.dynmat.out
```

The input file is Si.dynmat.in :

```
&input
fildyn = 'Si.dyn',
asr = 'simple',
/
```

fildyn: File containing the dynamical matrix

asr: A way to impose the acoustic sum rule (simple, crystal, one-dim, zero-dim, all)

Phonons at Γ in non-polar materials

The program dynmat.x produces the file **dynmat.out** which contains the new acoustic frequencies, which are exactly equal to zero

```
q =       0.0000      0.0000      0.0000
**************************************************************************
     freq (    1) =       0.000000 [THz] =       0.000000 [cm-1]
 ( 0.000000   0.000000     0.707107   0.000000     0.000000   0.000000   )
 ( 0.000000   0.000000     0.707107   0.000000     0.000000   0.000000   )
     freq (    2) =       0.000000 [THz] =       0.000000 [cm-1]
 (-0.707107   0.000000     0.000000   0.000000     0.000000   0.000000   )
 (-0.707107   0.000000     0.000000   0.000000     0.000000   0.000000   )
     freq (    3) =       0.000000 [THz] =       0.000000 [cm-1]
 ( 0.000000   0.000000     0.000000   0.000000     0.707107   0.000000   )
 ( 0.000000   0.000000     0.000000   0.000000     0.707107   0.000000   )
     freq (    4) =      15.302447 [THz] =     510.434704 [cm-1]
 ( 0.000000   0.000000     0.707107   0.000000     0.000000   0.000000   )
 ( 0.000000   0.000000    -0.707107   0.000000     0.000000   0.000000   )
     freq (    5) =      15.302447 [THz] =     510.434704 [cm-1]
 ( 0.000000   0.000000     0.000000   0.000000    -0.707107   0.000000   )
 ( 0.000000   0.000000     0.000000   0.000000     0.707107   0.000000   )
     freq (    6) =      15.302447 [THz] =     510.434704 [cm-1]
 (-0.707107   0.000000     0.000000   0.000000     0.000000   0.000000   )
 ( 0.707107   0.000000     0.000000   0.000000     0.000000   0.000000   )
**************************************************************************
~
```

Exercise 1.b:

Phonons dispersion in non-polar materials

Phonon dispersion in non-polar materials

Go to the directory with the input files:

cd into 01_Si_QE_b

In this directory you will find:

- *README.md* – File describing how to do the exercise
- *Si.scf.in* – Input file for the SCF ground-state calculation
- *Si.ph.in* – Input file for the phonon calculation at Γ
- *Si.q2r.in* – Input file for calculation of Interatomic Force Constants
- *Si.matdyn.in* – Input file for Fourier Interpolation for various q points
- *Si.plotband.in* - Input file for plotting a phonon dispersion
- **reference** – Directory with the reference results

Phonon dispersion in non-polar materials

Step 1: Perform a Self-Consistent Field ground-state calculation for silicon at the equilibrium structure using the pw.x program.

```
mpirun −np 2 pw.x −in Si.scf.in > Si.scf.out
```

Step 2: Perform a phonon calculation on a uniform grid of q points the ph.x program.

```
mpirun −np 2 ph.x −in Si.ph.in > Si.ph.out
```

```
&inputph
 prefix='Si',
 tr2_ph = 1.0d-14,
 amass(1) = 28.0855,
 ldisp = .true.,
 nq1 = 4,
 nq2 = 4,
 nq3 = 4,
 outdir='./out'
 fildyn='Si.dyn',
```

Flags for the calculation on a grid
Uniform grid of q points:

$$q_{ijk} = \frac{i-1}{nq1}\boldsymbol{G_2} + \frac{j-1}{nq2}\boldsymbol{G_1} + \frac{k-1}{nq3}\boldsymbol{G_1}$$

Phonon dispersion in non-polar materials

4x4x4 = 64 **q**-points => **Use of symmetry** => 8 non-equivalent **q** points

The file Si.dyn0 contains a list of the non-equivalent q points (8, in this case).

```
  4   4   4
  8
  0.000000000000000E+00   0.000000000000000E+00   0.000000000000000E+00
 -0.250000000000000E+00   0.250000000000000E+00  -0.250000000000000E+00
  0.500000000000000E+00  -0.500000000000000E+00   0.500000000000000E+00
  0.000000000000000E+00   0.500000000000000E+00   0.000000000000000E+00
  0.750000000000000E+00  -0.250000000000000E+00   0.750000000000000E+00
  0.500000000000000E+00   0.000000000000000E+00   0.500000000000000E+00
  0.000000000000000E+00  -0.100000000000000E+01   0.000000000000000E+00
 -0.500000000000000E+00  -0.100000000000000E+01   0.000000000000000E+00
```

The phonon code ph.x generates a file for every non-equivalent q point (Si.dyn1, Si.dyn2, ..., Si.dyn8), which contain information about dynamical matrices, phonon frequencies and atomic displacements

Phonon dispersion in non-polar materials

Step 3: Calculation of the Interatomic Force Constants (IFC) using the q2r.x program

IFCs in reciprocal space (Fourier transform of IFCs):

$$\tilde{C}_{s\alpha,s'\beta}(\boldsymbol{q}) = \frac{\partial^2 \mathrm{E}_{\mathrm{tot}}}{\partial \tilde{\boldsymbol{u}}_{s\alpha}(\boldsymbol{q}) \partial \tilde{\boldsymbol{u}}_{\beta}(\boldsymbol{q})}$$

where α, β are the cartesina direction and s,s' the atomic index

$$C_{s\alpha,s'\beta}(\boldsymbol{R}) = \sum_{\boldsymbol{q_n}} \tilde{C}_{s\alpha,s'\beta}(\boldsymbol{q}) e^{i\boldsymbol{q_n}(\boldsymbol{R}-\boldsymbol{R'})}$$

$$\tilde{C}_{s\alpha,s'\beta}(\boldsymbol{q}) \longrightarrow C_{s\alpha,s'\beta}(\boldsymbol{R})$$

Fourier transforms of IFCs on a grid of **q** points nq1 x nq2 x nq3 in reciprocal space

IFCs in a supercell nq1 x nq2 x nq3 in real space

Phonon dispersion in non-polar materials

Step 3: Calculation of the Interatomic Force Constants (IFC) using the q2r.x program

```
mpirun -np 2 q2r.x -in Si.q2r.in > Si.q2r.out
```

```
&input
fildyn = 'Si.dyn',
zasr = 'simple',
flfrc = 'Si444.fc',
/
```

- fildyn: File containing the dynamical matrix
- zasr: A way to impose the acoustic sum rule (simple, crystal, one-dim, zero-dim, all)
- flfrc: Output file of the interatomic force constants

Note: The denser the grid of q points, the larger the vectors R for which the IFCs are calculated.

Phonon dispersion in non-polar materials

Step 4: Calculate phonons at generic **q**' points using IFC using the matdyn.x code

$$\tilde{C}_{s\alpha,s'\beta}(\boldsymbol{q}) = \sum_{\boldsymbol{R}} C_{s\alpha,s'\beta}(\boldsymbol{R}) e^{-i\boldsymbol{q}'\boldsymbol{R}}$$

$$C_{s\alpha,s'\beta}(\boldsymbol{R}) \xrightarrow{\text{Fourier interpolation}} \tilde{C}_{s\alpha,s'\beta}(\boldsymbol{q}')$$

IFC's on a grid in real space

Fourier transforms of IFC's at generic q' points in reciprocal space

```
&input
asr = 'simple',
amass(1) = 28.0855,
flfrc = 'Si444.fc',
flfrq = 'Si.freq'
/
396
0.000000 0.000000 0.000000 0.000000
0.012658 0.000000 0.000000 0.012658...
```

```
mpirun –np 4 matdyn.x –in Si.matdyn.in > Si.matdyn.out
```

- flfrc: File containing the the interatomic force constants from q2r.x
- flfrq: Output file with the interpolated frequencies
- 396: Number of q points
- Coordinates of the q points

Phonon dispersion in non-polar materials

Step 5: Plot the phonon dispersion using the plotband.x program and gnuplot.

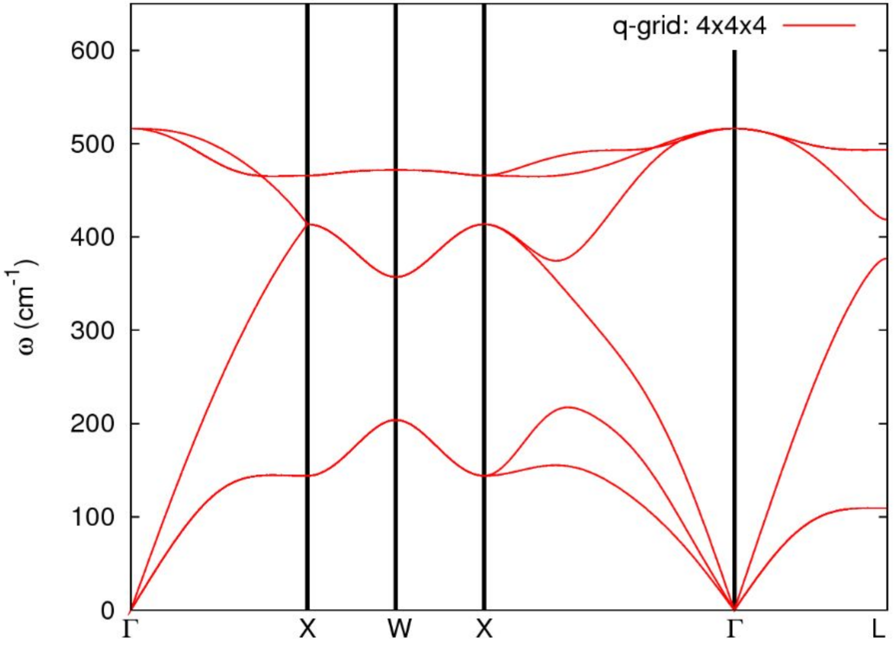
```
plotband.x < Si.plotband.in > Si.plotband.out
```

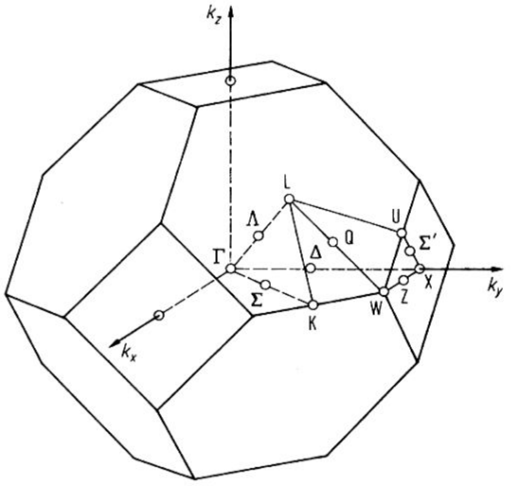
Si.freq	Input file with the frequencies at various q' points
0 600	Range of frequencies for a visualization
freq.plot	Output file with frequencies which will be used for plot
freq.ps	Plot of the dispersion (we will produce another one)
0.0	Fermi level (needed only for band structure plot)
100.0 0.0	Freq. step and reference freq. on the plot freq.ps

Use gnuplot and the file **plot_dispersion.gp** in order to plot the phonon dispersion of silicon (look at the file experimental_data.dat for the experimental reference).
You will get a postscript file phonon_dispersion.eps which you can visualize.

Phonon dispersion in non-polar materials

Phonon dispersion of silicon along some high-symmetry directions in the Brillouin zone (file phonon_dispersion.eps).





Phonon dispersion in non-polar materials

How to determine whether the quality of the Fourier interpolation is satisfactory? Compare with the direct calculation (no interpolation)!

Homework-1: Perform a direct phonon calculation (no interpolation) at several **q**' points and make a comparison with the phonon frequencies obtained from the interpolation. (Use exercise2 as an example).
Some **q**' points along the Gamma-X high symmetry line are listed in the file reference/q_points_direct_calc.txt

Homework-2: Perform a phonon dispersion calculation for several q-points grids (eg. 2x2x2, 4x4x4, and 6x6x6) and compare the dispersions. Do they converge?

Phonon dispersion in non-polar materials

The phonon modes are not always easy to visualize, especially if we are not at Γ. An online phonon visualizer is very helpful in this regard.

https://interactivephonon.materialscloud.io

Let’s click on the link and look at your phonons.

Exercise 2.a:

Phonons at Gamma in polar materials

Phonons at Γ in polar materials

Polar materials in the q = 0 limit: a macroscopic electric field appears as a consequence of the long-range character of the Coulomb interaction (incompatible with Periodic Boundary Conditions).
A non-analytic term must be added to Interatomic Force Constants at **q** = 0:

$$\tilde{C}_{s\alpha,s'\beta}(\boldsymbol{q}) = \tilde{C}^{analytic}_{s\alpha,s'\beta} + \frac{4\pi}{V}\frac{(\boldsymbol{q}\cdot\boldsymbol{Z}^*_s)_\alpha(\boldsymbol{q}\cdot\boldsymbol{Z}^*_{s'})_\beta}{\boldsymbol{q}\cdot\varepsilon_\infty\cdot\boldsymbol{q}}$$

Effective charges Z^*_s are related to polarization P induced by a lattice

$$Z^*_{s,\alpha\beta} = V\frac{\partial P_\alpha}{\partial u^\beta_s}$$

Dielectric tensor $\varepsilon^{\alpha\beta}_\infty$ is related to polarization P induced by an electric field E:

$$\varepsilon^{\alpha\beta}_\infty = \delta_{\alpha\beta} + 4\pi\left.\frac{\partial P_\alpha}{\partial E_\beta}\right|_{u_s(\boldsymbol{q}=0)=0}$$

All of the above can be calculated from (mixed) second order derivatives of the total energy.

Phonons at Γ in polar materials

Go to the directory with the input files:

cd into 02_AlAs_QE_b

In this directory you will find:

- *README.md* – File describing how to do the exercise
- *AlAs.scf.in* – Input file for the SCF ground-state calculation
- *AlAs.ph.in* – Input file for the phonon calculation at Γ
- *AlAs.dynmat.in* – Input file for Fourier Interpolation for various q points
- ***reference*** – Directory with the reference results

Phonons at Γ in polar materials

Step 1: Perform a Self-Consistent Field ground-state calculation for AsAl at the equilibrium structure using the pw.x program.

```
mpirun –np 2 pw.x –in AlAs.scf.in > AlAs.scf.out
```

Step 2: Perform a phonon calculation in Gamma using ph.x.

```
mpirun –np 2 ph.x –in AlAs.ph.in > AlAs.ph.out
```

```
Phonons at Gamma
&inputph
prefix = 'AlAs',
tr2_ph = 1.0d-14,
amass(1) = 26.98,
amass(2) = 74.92,
epsil = .true.
outdir = './tmp'
fildyn = 'AlAs.dyn',
/
0.0 0.0 0.0
```

Phonons at Γ in polar materials

In the file ph.AlAs.out you will find information about the dielectric tensor and effective charges (BECs)

```
     Effective charges (d Force / dE) in cartesian axis with asr applied:
      atom      1   Al Mean Z*:       4.43445
 E*x (        4.43445         0.00000        -0.00000 )
 E*y (        0.00000         4.43445        -0.00000 )
 E*z (       -0.00000        -0.00000         4.43445 )
      atom      2   As Mean Z*:      -4.43445
 E*x (       -4.43445        -0.00000         0.00000 )
 E*y (       -0.00000        -4.43445         0.00000 )
 E*z (        0.00000         0.00000        -4.43445 )

 Diagonalizing the dynamical matrix

 q = (    0.000000000   0.000000000   0.000000000 )

**************************************************************************
 freq (    1) =      -0.062455 [THz] =      -2.083289 [cm-1]
 freq (    2) =      -0.062455 [THz] =      -2.083289 [cm-1]
 freq (    3) =      -0.062455 [THz] =      -2.083289 [cm-1]
 freq (    4) =      11.807415 [THz] =     393.852972 [cm-1]
 freq (    5) =      11.807415 [THz] =     393.852972 [cm-1]
 freq (    6) =      11.807415 [THz] =     393.852972 [cm-1]
**************************************************************************
```

$$Z^*_{s,\alpha\beta} = \frac{V}{e}\frac{\partial P_\alpha}{\partial u_{s,\beta}(\boldsymbol{q}=0)} = \frac{1}{e}\frac{\partial F_{s\beta}}{\partial \mathcal{E}_\alpha(\boldsymbol{q}=0)}$$

No LO-TO splitting

Phonons at Γ in polar materials

Step 3: Impose Acoustic Sum Rule and add the non analytic LO-TO splitting using the dynmat.x program

```
mpirun –np 2 dynmat.x –in AlAs.dynmat.in > AlAs.dynmat.out
```

```
&input
  fildyn = 'AlAs.dyn',
  asr='simple',
  amass(1)=26.98,
  amass(2)=74.92
  q(1) = 1.0,
  q(2) = 0.0,
  q(3) = 0.0
/
```

Direction in the Brillouin zone along which we want to compute the LO-TO splitting

```
                  IR activities are in (D/A)^2/amu units

# mode   [cm-1]    [THz]      IR
    1     -0.00   -0.0000    0.0000
    2      0.00    0.0000    0.0000
    3      0.00    0.0000    0.0000
    4    393.86   11.8075   22.8709
    5    393.86   11.8075   22.8709
    6    430.19   12.8969   22.8709
```

LO-TO splitting

Phonons at Γ in polar materials

Step 3: Impose Acoustic Sum Rule and add the non analytic LO-TO splitting using the dynmat.x program

```
mpirun −np 2 dynmat.x −in AlAs.dynmat.in > AlAs.dynmat.out
```

```
&input
  fildyn = 'AlAs.dyn',
  asr='simple',
  amass(1)=26.98,
  amass(2)=74.92
  q(1) = 1.0,
  q(2) = 0.0,
  q(3) = 0.0
/
```

from effective charges

IR activities are in (D/A)^2/amu units

# mode	[cm-1]	[THz]	IR
1	-0.00	-0.0000	0.0000
2	0.00	0.0000	0.0000
3	0.00	0.0000	0.0000
4	393.86	11.8075	22.8709
5	393.86	11.8075	22.8709
6	430.19	12.8969	22.8709

IR intensities

$$I_{IR}(\nu) = \sum_{\alpha} \left| \sum_{s\beta} Z^{*\alpha\beta}_{s} u^{\beta}_{s}(\nu) \right|^2$$

can be calculated directly from effective charges and phonon displacement patterns

Exercise 2.b:

Phonons dispersion in non-polar materials

Phonon dispersion in non-polar materials

Go to the directory with the input files:

cd into 02_AlAs_QE_b

In this directory you will find:

- *README.md* – File describing how to do the exercise
- *AlAs.scf.in* – Input file for the SCF ground-state calculation
- *AlAs.ph.in* – Input file for the phonon calculation at Γ
- *AlAs.q2r.in* – Input file for calculation of Interatomic Force Constants
- *AlAs.matdyn.in* – Input file for Fourier Interpolation for various q points
- *AlAs.plotband.in* – Input file for plotting a phonon dispersion
- ***reference*** – Directory with the reference results

Phonon dispersion in non-polar materials

Step 1: Perform a Self-Consistent Field ground-state calculation at the equilibrium structure using the pw.x program.

```
mpirun –np 2 pw.x –in AlAs.scf.in > Si.scf.out
```

Step 2: Perform a phonon calculation on a uniform grid of q points the ph.x program.

```
mpirun –np 2 ph.x –in AlAs.ph.in > AlAs.ph.out
```

Step 3: Calculation of the Interatomic Force Constants (IFC) using the q2r.x program

```
mpirun –np 2 q2r.x –in AlAs.q2r.in > AlAs.q2r.out
```

Step 4: Calculate phonons at generic **q**' points using IFC using the matdyn.x code

```
mpirun –np 2 matdyn.x –in AlAs.matdyn.in > AlAs.matdyn.out
```

Step 5: Plot the phonon dispersion using the plotband.x program and gnuplot.

```
plotband.x < AlAs.plotband.in > AlAs.plotband.out
```

Exercise 3:

Phonons and forces with machine learning potentials

Phonons and forces with machine learning potentials

ADVANTAGES

PERTURBATION THEORY

very efficient for primitive cells
no supercell needed
no need to choose the displacement

FINITE DIFFERENCES

very general
can handle large primitive cells

→ **use in conjunction with machine learning interatomic potentials**

UPET
UNIVERSAL ATOMISTIC MODELS

Phonons and forces with machine learning potentials

Step 1: Open the Jupyter-Lab (icon on desktop). Open the notebook with kernel metatomic
Step 2: Energy dependence of the volume and relaxation with ML forces
Step 3: Phonons computed with finite-differences

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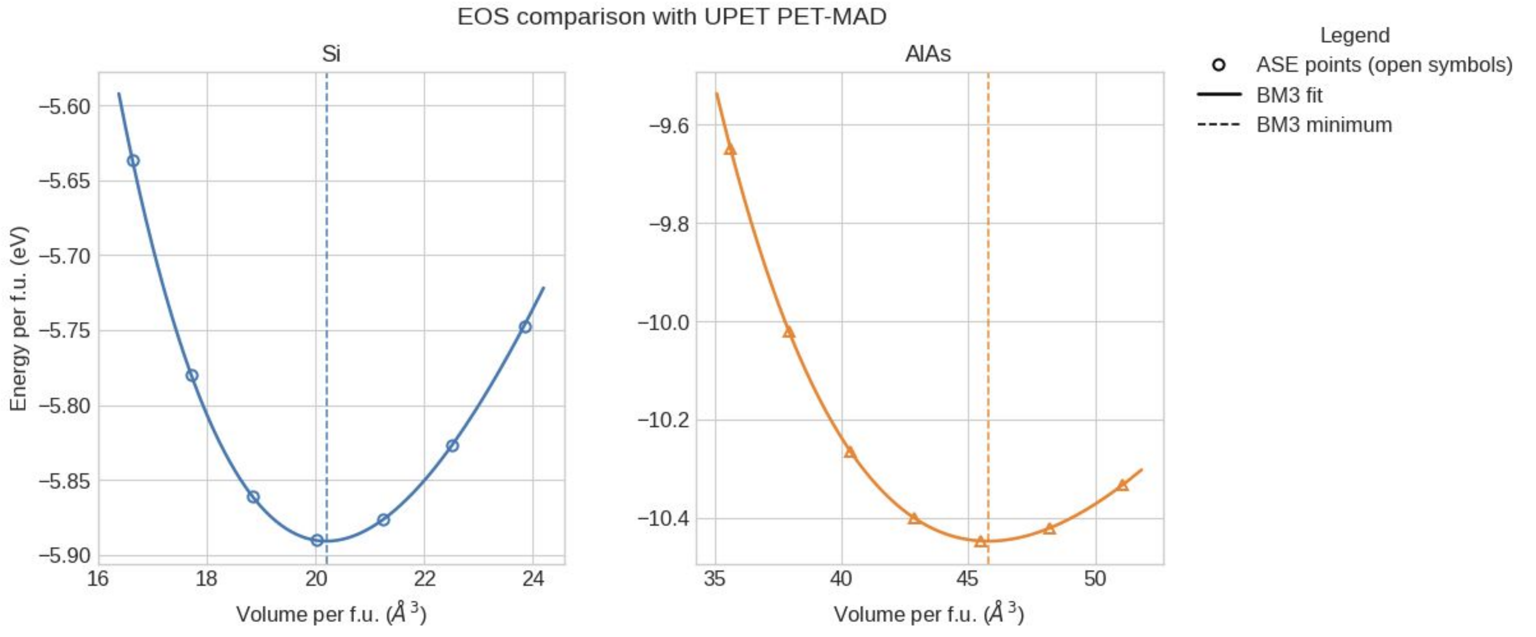
UNIVERSAL ATOMISTIC MODELS

```
cd 03_phonons_mlip
jupyter-lab 03_phonons_mlip/day2_mlip_ase_eos_relax_phonons.ipynb
```



Thank you!

Questions?